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SUPERSEDING
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MILITARY SPECIFICATION

INCLINOMETER, BALL STYLE

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of ball-style inclinometers for use in military ground-based vehicles.

1.2 Classification. Inclinometers shall be of the types specified in table I.

TABLE I. Classification of inclinometers.

Type	Scale degrees
I	0 to 20°
II	0 to 10°

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 6695

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810D	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-45662	- Calibration Systems Requirements.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish inclinometers which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information, shall be representative of the units to be furnished to the Government. All subsequent inclinometers delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Materials utilized in the manufacture of the inclinometer shall be entirely adequate for the intended purpose under the performance and environmental conditions imposed by this specification without any degradation in performance (see 4.6.1 and 6.3).

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3.2.1 Metals and finishes. All metals shall be corrosion-resistant type or suitably protected to resist corrosion during the performance and environmental conditions specified herein. Dissimilar metals shall be in accordance with MIL-STD-454, requirement 16 (see 4.6.1).

3.2.2 Nonmetals. All nonmetals shall be compatible (see 4.6.1).

3.2.3 Damping fluid. The damping fluid utilized shall have properties which will enable the inclinometer to meet the performance and environmental requirements herein and shall be sufficiently colorless not to interfere with satisfactory viewing of the ball (see 4.6.1).

3.3 Design and construction. The assembly shall be constructed so that no parts will work loose in service and shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, service, and environment. Processes and finishes utilized in the manufacture of the inclinometer shall be entirely adequate for the intended purpose under the performance and environmental conditions imposed by this specification without any degradation in performance (see 4.6.1).

3.3.1 Configuration. Inclinometers shall be designed and constructed in accordance with the applicable figure specified in 3.3.1.1 or 3.3.1.2 (see 4.6.2).

3.3.1.1 Type I. See figure 1.

3.3.1.2 Type II. See figure 2.

3.3.2 Component parts.

3.3.2.1 Ball. The ball shall be made of black glass or similar material. It shall be highly polished and accurately spherical (see 4.6.1).

3.3.2.1.1 Ball diameter.

3.3.2.1.1.1 Type I. The diameter of the ball shall be 0.359 ± 0.002 inch [9.11 ± 0.05 millimeters (mm)] (see 4.6.2).

3.3.2.1.1.2 Type II. The diameter of the ball shall be 0.234 ± 0.063 inch (5.94 ± 1.60 mm) (see 4.6.2).

3.3.2.2 Tube. The glass tube shall be made of clear, annealed glass tubing, free from defects which may interfere with normal reading of the inclinometer. The inside of the tube shall be smooth and uniform in order to ensure that the ball will roll smoothly. The tube shall be firmly secured in the frame by means of a filler. An expansion chamber shall be provided at each end of the tube (see 4.6.1).

3.3.2.2.1 Wall thickness.

3.3.2.2.1.1 Type I. The wall thickness shall be not less than 0.0469 inch (1.19 mm) (see 4.6.2).

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3.3.2.2.1.2 Type II. The wall thickness shall be not less than 0.0390 inch (0.99 mm) (see 4.6.2).

3.3.2.3 Frame. The frame shall be fabricated of aluminum alloy (see 4.6.1).

3.3.3 Weight. The weight of the inclinometer shall not exceed 0.3 pound [0.14 kilogram (kg)] (see 4.6.2).

3.4 Performance.

3.4.1 Air bubble visibility. The air bubble shall not be visible when the inclinometer is viewed from a position 12 inches (300 mm) directly in front of the inclinometer zero mark with the inclinometer inclined to 20 degrees [350 milliradians (mrad)] for type I and 10 degrees (175 mrad) for type II (see 4.6.3).

3.4.2 Scale error. Scale error of the inclinometer shall not exceed the tolerance at degrees of inclination specified in table II (see 4.6.4).

TABLE II. Tolerance for degree of inclination.

Inclination		Tolerance	
Type I degrees (mrad)	Type II degrees (mrad)	Type I degrees (mrad)	Type II degrees (mrad)
0 (0)	0 (0)	1.0 (18)	1.0 (18)
5 (88)	5 (88)	1.5 (26)	1.5 (26)
10 (175)	10 (175)	2.0 (35)	2.0 (35)
15 (263)	- -	2.5 (44)	- -
20 (350)	- -	2.5 (44)	- -

3.4.3 Ball visibility. With the inclinometer tipped in such a manner that the ball is at rest at either end of the tube, not less than half of the ball shall be visible when viewed from a position 12 inches (300 mm) directly in front of the zero mark of the inclinometer (see 4.6.5).

3.4.4 Damping.

3.4.4.1 Room temperature. The time of ball roll from the zero mark to 20 degrees (350 mrad) for type I, or 10 degrees (175 mrad) for type II, shall not exceed 0.5 second at room ambient temperature (see 4.6.6).

3.4.4.2 Low temperature. The time of ball roll from the zero mark to 20 degrees for type I, or 10 degrees for type II, shall not exceed 10 seconds at -26 degrees Fahrenheit (°F) [-32 degrees Celsius(°C)] (see 4.6.6).

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3.5 Environmental.

3.5.1 Temperatures.

3.5.1.1 High temperature (operating). The inclinometer shall meet the performance requirements specified herein during and after exposure to ambient temperatures up to 140°F (60°C) (see 4.7.1.1).

3.5.1.2 High temperature (nonoperating). The inclinometer shall not be damaged and shall meet the performance requirements specified herein after exposure to temperatures up to 158°F (70°C) (see 4.7.1.2).

3.5.1.3 Low temperature (operating). The inclinometer shall meet the performance requirements specified herein during and after exposure to ambient temperatures down to -26°F (-32°C) (see 4.7.1.3).

3.5.1.4 Low temperature (nonoperating). The inclinometer shall not be damaged and shall meet the performance requirements specified herein after exposure to temperatures down to -65°F (-54°C) (see 4.7.1.4).

3.5.2 Leakage. The size of the air bubble shall show no appreciable change after exposure to a temperature of 140°F (60°C) (see 4.7.2).

3.5.3 Relative humidity (nonoperating). The inclinometer shall not be damaged and shall meet the performance requirement specified herein after exposure to relative humidity ranges to 100 percent at 86°F (30°C) and 2 percent at 155°F (68°C) (see 4.7.3).

3.5.4 Atmospheric pressure.

3.5.4.1 Operating. The inclinometer shall meet the performance requirements during and after exposure to an atmospheric pressure equivalent to an altitude of 8000 feet [2400 meters (m)] (see 4.7.4.1).

3.5.4.2 Nonoperating. The inclinometer shall not be damaged and shall meet the performance requirements specified herein after exposure to an atmospheric pressure equivalent to an altitude of 40 000 feet (12 000 m) (see 4.7.4.2).

3.5.5 Salt fog (nonoperating). The inclinometer shall not be damaged and shall meet the performance requirements specified herein after exposure to salt fog (see 4.7.5).

3.5.6 Fungus resistance. The inclinometer shall show no evidence of fungus growth and shall meet the performance requirements specified herein after exposure to conditions favorable to fungus growth (see 4.7.6).

3.5.7 Shock (nonoperating). The inclinometer shall not be damaged and shall meet the performance requirements specified herein after exposure to half-sine wave shock impulses of 25 gravity units (g) applied along three mutually perpendicular axes for 11 milliseconds (ms) (see 4.7.7).

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3.5.8 Vibration (nonoperating). The inclinometer shall not be damaged and shall meet the performance requirements specified herein after exposure to simple harmonic motion applied along three mutually perpendicular axes at peak acceleration of ± 10 g at all frequencies within a range of 55 to 2000 hertz (Hz) for type I, and a range of 20 to 2000 Hz for type II. Frequency sweep time shall be logarithmic and sweep time shall be 35 ± 5 minutes. In addition the inclinometer shall withstand 5 minutes vibration at each observed resonant frequency (see 4.7.8).

3.6 Marking. All markings shall be durable and permanent to withstand usage encountered in service. The inclinometer shall be marked in accordance with MIL-STD-130 with the following information (see 4.6.2):

a. Manufacturers name or code.

b. Operation markings:
Type 1 - see figure 1.
Type II - see figure 2.

c. Manufacturers identifying number (optional).

d. Date code (month /year).

e. US Army Tank Automotive Command part number:

Type 1 - 12265593.
Type 2 - 12265594.

f. Design activity identification number: 19207.

3.7 Workmanship. All parts shall be fabricated and assembled in a thorough workmanship-like manner to applicable drawing requirements. They shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, corrosion products or other foreign matter that would affect appearance and performance. All required marking shall be neat, sharply defined, permanent and in accordance with drawing requirements (see 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply wih all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection.

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspection (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under ambient conditions:

a. Air temperature 73 $\pm$ 18°F (23 $\pm$ 10°C)
b. Barometric pressure 28.5^{+2}_{-3} inches (725^{+50}_{-75}mm) mercury (Hg)
c. Relative humidity 50 $\pm$ 30 percent

4.4 First article inspection. First article inspection shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply inclinometers that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two inclinometers of the specified type. Preproduction inspection shall consist of inspection as specified in table III.

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4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two inclinometers from the first 20 of the specified type produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table III.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance	
				Examination	Test
Materials and construction	3.2 thru 3.3 3.3.2.1, 3.3.2.2 and 3.3.2.3	4.6.1	X		X
Defects (see 4.5.2 and Table IV)	3.3.1, 3.3.2.1.1, 3.3.3, 3.3.2.2.1, 3.6 and 3.7	4.6.2	X	X	
Air bubble visibility	3.4.1	4.6.3	X		X
Scale error	3.4.2	4.6.4	X		X
Ball visibility	3.4.3	4.6.5	X		X
Damping	3.4.4				
Room temperature	3.4.4.1	4.6.6	X		X
Low temperature	3.4.4.2	4.6.6	X		
Temperatures	3.5.1				
High (operating)	3.5.1.1	4.7.1.1	X		
High (nonoperating)	3.5.1.2	4.7.1.2	X		
Low (operating)	3.5.1.3	4.7.1.3	X		
Low (nonoperating)	3.5.1.4	4.7.1.4	X		
Leakage	3.5.2	4.7.2	X		
Relative humidity (nonoperating)	3.5.3	4.7.3	X		
Atmospheric pressure	3.5.4				
(operating)	3.5.4.1	4.7.4.1	X		
(nonoperating)	3.5.4.2	4.7.4.2	X		
Salt fog (nonoperating)	3.5.5	4.7.5	X		
Fungus resistance	3.5.6	4.7.6	X		
Shock (nonoperating)	3.5.7	4.7.7	X		
Vibration (nonoperating)	3.5.8	4.7.8	X		

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4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article test shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the inclinometers of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least five consecutively produced lots that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQL's.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQLs):

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table IV.

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TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	NONE	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability, out of tolerance (see 3.3.1, 3.3.2.1.1 and 3.3.2.2.1).	SIE 1/
102	Improper weight (see 3.3.3).	SIE
103	Improper marking (see 3.6).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1, 3.3.2.1.1 and 3.3.2.2.1).	SIE
202	Faulty workmanship (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment

4.5.3 Tests. Each inclinometer shall be subjected to the tests specified in table III.

4.5.4 Failure. Failure of any inclinometer assemblies to pass any of the specified quality conformance tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2 through 3.3, 3.3.2.1, 3.3.2.2 and 3.3.2.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3.1, 3.3.2.1.1, 3.3.2.2.1, 3.3.3, 3.6 and 3.7 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

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4.6.3 Air bubble visibility. To determine conformance to 3.4.1, the inclinometer shall be tipped in such a manner that all air in the tube is trapped in one end of the tube. Then with the back of the inclinometer in a vertical plane and with the line joining the center of the mounting holes inclined to the horizontal at an angle of 20 degrees (350 mrad) for type I and 10 degrees (175 mrad) for type II, and with the end of the tube containing the bubble downward, no part of the air bubble shall be visible when the instrument is viewed from a position 12 inches (300 mm) directly in front of the zero mark of the inclinometer.

4.6.4 Scale error. To determine conformance to 3.4.2, the inclinometer shall be tested for scale errors at the angles of inclination specified in table II. All scale errors shall be determined with the back of the instrument in a vertical plane and with gentle tapping of the instrument before a test reading is taken. The error at the zero point of the scale shall be determined with the line joining the center of the mounting holes inclined to the horizontal at the specified angles. The ball shall roll smoothly throughout the length of the tube included between the degree marks when the inclinometer is slowly inclined to the horizontal with the back of the instrument in a vertical plane, and with gentle tapping of the inclinometer. The ball shall not stick at either end of the tube.

4.6.5 Ball visibility. To determine conformance to 3.4.3, tip the inclinometer so that the ball is at rest at either end of the tube. Not less than half of the ball shall be visible when viewed from a position 12 inches [30.5 centimeters (cm)] directly in front of the zero mark of the inclinometer.

4.6.6 Damping. To determine conformance to 3.4.4.1 and 3.4.4.2, with the back of the inclinometer in a vertical plane and with the line joining the centers of the mounting holes inclined to the horizontal at an angle of 24 degrees (419 mrad) for type I or 15 degrees (263 mrad) for type II, tip the inclinometer suddenly through a vertical angle of 48 degrees (838 mrad) for type I, or 30 (526 mrad) degrees for type II in order to ensure that the ball will roll to the opposite end of the tube. The elapsed time to roll the ball from the zero mark to the maximum degree mark of the inclinometer for type I or type II shall not exceed the time specified in 3.4.4.

4.7 Environmental.

4.7.1 Temperatures.

4.7.1.1 High temperature (operating). To determine conformance to 3.5.1.1, the inclinometer shall be tested in accordance with MIL-STD-810D, method 501.2, procedure II. The temperature shall be 140°F (60°C). The inclinometer shall pass the tests of 4.6.3 through 4.6.6 during and after exposure.

4.7.1.2 High (nonoperating). To determine conformance to 3.5.1.2, the inclinometer shall be tested in accordance with MIL-STD-810D, method 501.2, procedure I. The temperature shall be 158°F (70°C). Subsequently, the inclinometer shall pass the tests of 4.6.3 through 4.6.6.

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4.7.1.3 Low (operating). To determine conformance to 3.5.1.3, the inclinometer shall be tested in accordance with MIL-STD-810D, method 502.2, procedure II. The temperature shall be -26°F (-32°C). The inclinometer shall pass the tests of 4.6.3 through 4.6.6 during and after exposure.

4.7.1.4 Low (nonoperating). To determine conformance to 3.5.1.4, the inclinometer shall be tested in accordance with MIL-STD-810D, method 502.2, procedure I. The temperature shall be -65°F (-54°C). Subsequently, the inclinometer shall pass the tests of 4.6.3 through 4.6.6.

4.7.2 Leakage. To determine conformance to 3.5.2, the size of the air bubble in the inclinometer shall be determined with the instrument at room temperature. The inclinometer shall then be maintained at a temperature of 140°F (60°C) for a period of 1 hour. The size of the air bubble shall then be determined with the inclinometer at room temperature. There shall be no appreciable change in the size of the air bubble before and after the inclinometer has been subjected to the temperature specified.

4.7.3 Relative humidity (nonoperating). To determine conformance to 3.5.3, the inclinometer shall be tested in accordance with MIL-STD-810D, method 507.2, procedure I. Subsequently, the inclinometer shall pass the tests of 4.6.3 through 4.6.6.

4.7.4 Atmospheric pressure.

4.7.4.1 Operating. To determine conformance to 3.5.4.1, the inclinometer shall be tested in a chamber stabilized at a temperature of 126°F (52°C) and a pressure equivalent to an altitude of 8000 feet (2400 m) for 8 hours with no change in temperature or pressure. The inclinometer shall pass the tests of 4.6.3 through 4.6.6 during and after exposure.

4.7.4.2 Nonoperating. To determine conformance to 3.5.4.2, the inclinometer shall be tested in accordance with MIL-STD-810D, method 500.2, procedure I including test for pressure equivalent to an altitude of 40 000 feet (12 000 m). Subsequently, the inclinometer shall pass the tests of 4.6.3 through 4.6.6.

4.7.5 Salt fog (nonoperating). To determine conformance to 3.5.5, the inclinometer shall be tested in accordance with MIL-STD-810D, method 509.2, procedure I. Subsequently, the inclinometer shall pass the tests of 4.6.3 through 4.6.6.

4.7.6 Fungus resistance. To determine conformance to 3.5.6, the inclinometer shall be tested in accordance with MIL-STD-810D, method 508.3. Subsequently, the inclinometer shall pass the tests of 4.6.3 through 4.6.6. In lieu of a test, the contractor shall maintain records providing proof or certification that the inclinometer is constructed of materials that will not support fungus growth.

4.7.7 Shock (nonoperating). To determine conformance to 3.5.7, the inclinometer shall be tested in accordance with MIL-STD-202, method 213B, condition J. A peak acceleration of 25g shall apply. Subsequently, the inclinometer shall pass the tests of 4.6.3 through 4.6.6.

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4.7.8 Vibration (nonoperating). To determine conformance to 3.5.8, the inclinometer shall be tested in accordance with MIL-STD-202, method 204C, condition C. Subsequently, the inclinometer shall pass the tests of 4.6.3 through 4.6.6.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The inclinometer is intended to be used in military ground vehicles to indicate the attitude of the vehicle with respect to the horizontal.

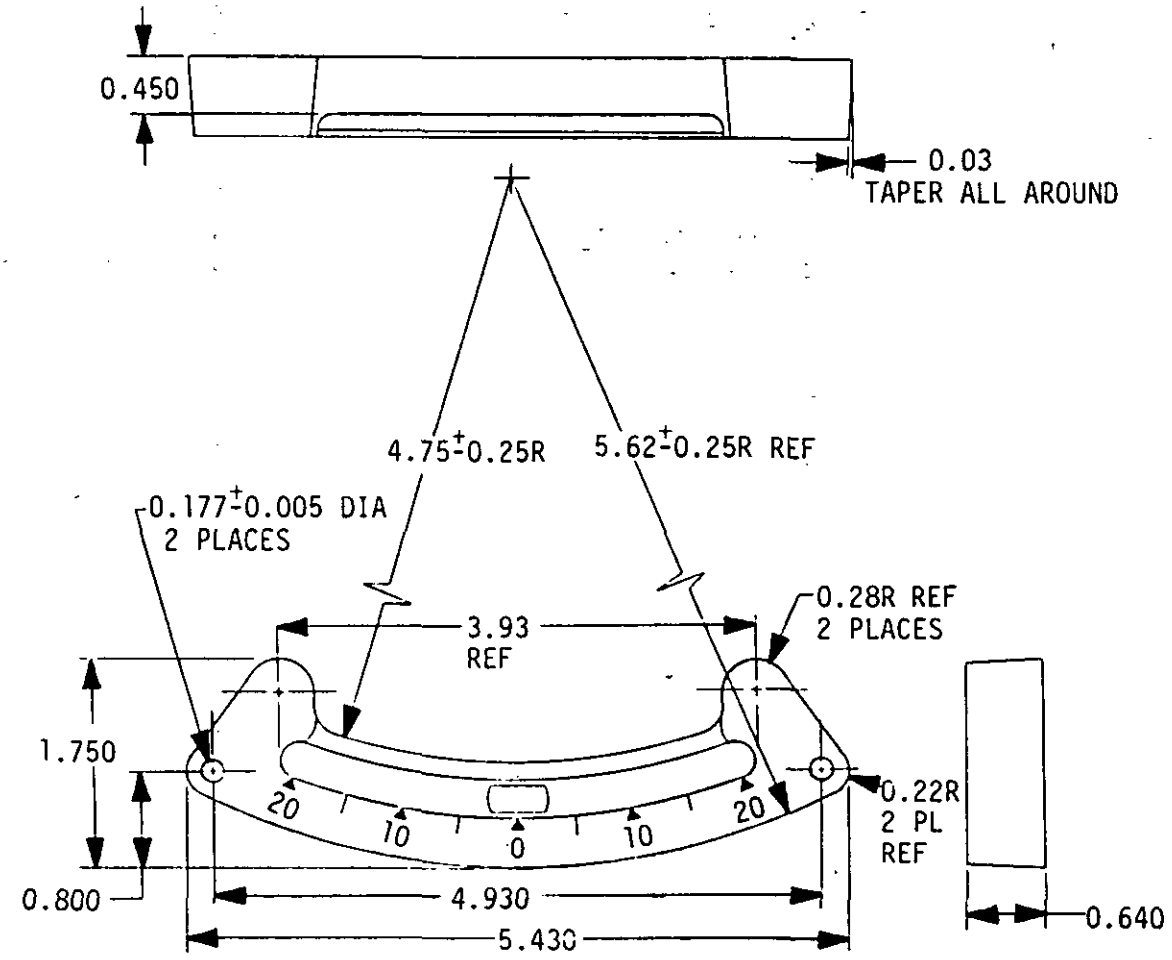
6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and nomenclature (see 1.2).
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If preproduction inspection is required (see 4.4.1).
h. If initial production inspection is not required (see 4.4.2).
i. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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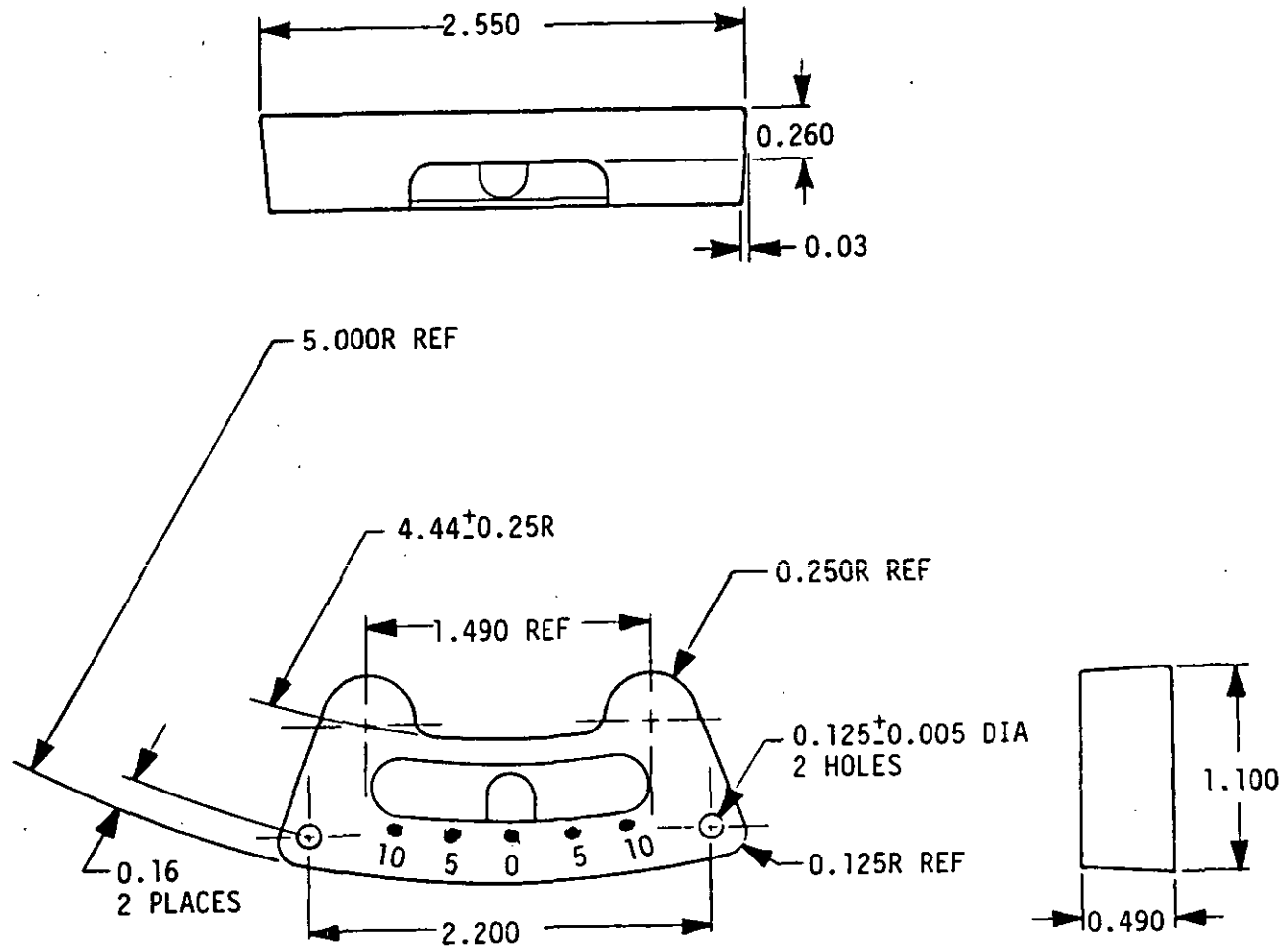


NOTES:
1. Unless otherwise noted tolerance is ±0.015.
2. Dimensions are in inches, but not to scale.

OPERATION MARKINGS			
MARKINGS	HEIGHTS OR LENGTH (INCHES)	WIDTH OF LINE OR GRADUATIONS (INCH ±0.005)	MATERIAL OR FINISH
NUMERALS	$0.171^{+0.000}_{-0.020}$	0.030	WHITE
5 AND 15 DEGREE LINES	$0.171^{+0.000}_{-0.020}$	0.030	FILLED
TRIANGLES AT 0°, 10°, AND 20°	0.093±0.010 INCH SIDES EQUILATERAL TRIANGLES		
REAR PORTION BEHIND GLASS TUBE			WHITE
EXTERIOR METALLIC SURFACES			DURABLE DULL BACK

FIGURE 1. Type 1 inclinometer.

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NOTE:
1. Dimensions are in inches.

OPERATION MARKINGS			
MARKINGS	HEIGHTS OR LENGTH (INCHES)	WIDTH OF LINE OR GRADUATIONS (INCH ± 0.005)	MATERIAL OR FINISH
NUMERALS	0.125±0.015	0.030	WHITE FILLED
FILLED CIRCLES AT 0°, 5°, AND 10°	$0.050^{+0.020}_{-0.010}$ FILLED CIRCLES		
REAR PORTION BEHIND GLASS TUBE			WHITE
EXTERIOR METALLIC SURFACES			DURABLE DULL BACK

FIGURE 2. Type II inclinometer.

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Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 6695-A069)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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